

Md. Ashfaul Haque

AI Product Lead & Senior ML Engineer || Enterprise AI,
Computer Vision & Edge Deployment

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Profile Summary

Senior AI/ML Engineer with **4+ years of experience** as **AI Product Owner** for enterprise AI solutions across banking, logistics, manufacturing, industrial safety, and surveillance. Defines **product vision** and owns the full **discovery to deployment lifecycle** covering requirement analysis, MVP scoping, AI feasibility assessment, data strategy, model development, edge deployment, KPI tracking, and post-launch improvement. Proven record of **aligning cross-functional teams** across **20+ AI detection scenarios** deployed on **1500+ live camera feeds in 700+ locations**. Brings **strategic product thinking**, **customer-centric discovery**, and **data-driven execution** to enterprise AI challenges across regulated, high-complexity environments.

Professional Experience

Bondstein Technologies Ltd. (RUNNER Group)

Senior Software Engineer (AI & ML) | AI Product Owner

Dhaka, Bangladesh

May 2022 – Present

- **Define product vision** and own end-to-end lifecycle of multiple enterprise AI products covering **product discovery**, requirement analysis, feasibility validation, data strategy, model development, deployment planning, monitoring, and continuous improvement.
- Being the **central coordination point** between business, client stakeholders, AI engineering, data annotation, UI/UX design, backend/frontend, hardware, and field deployment teams to align scope, milestones, risks, and deliverables.
- Lead and mentor a team of junior ML engineers through agile sprint cycles assigning tasks, reviewing model/code quality, managing **backlog prioritization**, tracking delivery KPIs, and guiding **research-to-production hand-off**.
- Structure use cases, on-site recce, and technical pitch validation for edge AI deployments; translate business needs and operational constraints into clear AI product requirements.
- Define practical **success criteria** for AI deployments including accuracy, latency, camera coverage, false-alert reduction, operational usability, scalability, and maintainability; monitor post-deployment **product adoption**, **customer satisfaction signals**, and performance trends to drive **continuous improvement**.
- Drive **responsible and scalable AI practices** by considering consent, data security, access control, model limitations, operational risks, and human-in-the-loop review for sensitive AI systems; ensure **documentation standards**, **compliance requirements**, and quality controls are maintained throughout the product lifecycle.

Selected Enterprise AI Products & Outcomes

- **AI Vision-Based Bank & ATM Monitoring Platform – Brac Bank:** Product and technical lead for launching and scaling live inference on **1000+ cameras across 650+ locations**; translated banking security needs into **12 AI detection scenarios** covering security events, behavioral identification, and real-time alerting workflows; **enabled the bank to scale security operations across branches without proportional headcount growth**, coordinating deployment planning and feedback loops with client and engineering teams.
- **Intrusion, Safety & SOP Monitoring Platform:** Led a centralized multi-camera industrial monitoring product covering **90+ cameras** for intrusion detection, virtual fencing, object tracking, PPE/harness detection, PIV detection, pathway monitoring, and speed monitoring; improved detection accuracy to **97–99%** and reduced dependency on onsite manual checking through automated alerting.
- **Central SOP Monitoring System:** Delivered computer-vision SOP monitoring across **17 locations and 150+ cameras** in Bangladesh, supporting gesture tracking, unique object tracking, multi-camera monitoring, day/night performance, and robust operation under varied lighting conditions; **proactively identified expansion opportunities** and grew product coverage by approximately **40% year-over-year** through iterative rollout and stakeholder feedback.
- **Container Management System:** Led delivery of an edge AI container management solution for depot operations with capacity of **6000+ containers** and **3000+ daily rolling movements**; scoped tracking requirements, deployment constraints, and edge execution using **NVIDIA Jetson Orin Nano**, reducing operational inefficiency by approximately **90%** and delivering **significant cost savings** through elimination of manual tracking and reconciliation processes.
- **Smart Parking & Bengali ANPR Solution:** Launched full ML backend for Bengali number-plate detection and recognition under day/night conditions; achieved **0.5–1 second ANPR** response time, handled **15,000+ annotations**, model fine-tuning, on-site deployment, and client-facing performance validation.

- **Deepfake Bengali Voice & Lipsync Production Pipeline:** Scaled an AI video production pipeline generating **2600+ videos** using custom Bengali voice generation and lipsync; managed quality controls, throughput optimization, and ethical use considerations for identity-based generative AI.
- **Crowd Activity Analytics Product:** Developed crowd activity monitoring, heatmap generation, and customizable analytics using Python backend and PySide/PyQt frontend to support client-specific operational insights.
- **Early-Stage AI Product Design & Feasibility (3 Initiatives):** Led **product scoping, solution architecture, and feasibility analysis** for multiple new-product initiatives — including BOPP/BOPET production-line visual QC targeting defect detection at 500–1200 RPM; AI-enabled inventory and delivery queue management covering product placement and forklift movement tracking; and Mobil engine oil product authentication with QR/visual verification and distribution traceability — each covering camera, lighting, AI inference design, and business case development.
- **TRP Monitoring & OCR/Card Interpretation:** Delivered end-to-end AI-enabled monitoring and interpretation tools including hardware planning, backend processing, Python GUI, packaging design, and parallel processing implementation.

Gunness and Associates

Researcher / Administrative Coordinator

Remote

May 2021 – Feb 2022

- Conducted immigration program research, NOC matching, lead management, client profiling, and client communication management.

China Harbour Engineering Company Ltd.

Assistant Service Engineer

Bangladesh

Apr 2018 – Mar 2019

- Supported installation, inspection, and documentation for electronic instruments, cable trays, junction boxes, control panels, marshalling cabinets, PLC wiring, and RFIs.

Product, AI & Technical Skills

Product Leadership:	Product vision, roadmap planning, MVP scoping , backlog prioritization, user stories & acceptance criteria , stakeholder alignment, agile delivery , market analysis , risk tracking, technical documentation.
AI Product Metrics:	Performance: model accuracy, precision/recall, false positive/negative rates, latency, edge throughput; Business: operational efficiency improvement, product adoption, rollout coverage growth, customer satisfaction, annotation quality.
AI/ML:	Computer Vision, Object Detection, Object Tracking, OCR, ANPR, Face Recognition, Deepfake/Lipsync, Voice Cloning, NLP, Predictive Maintenance, Federated Learning, Explainable AI, Transfer Learning, Few-shot / Siamese Learning.
Frameworks & Tools:	Python, PyTorch, TensorFlow/Keras, Scikit-learn, OpenCV, Ultralytics YOLO, DeepFace, Flask, FastAPI, PySide/PyQt, Pandas, NumPy, Matplotlib, LaTeX, Git, Postman.
Deployment:	NVIDIA Jetson Orin Nano, edge AI deployment, RTSP camera streams, multi-camera inference, Linux server setup, API integration, model optimization, real-time system design.

Education

Brac University

M.Sc. in Computer Science & Engineering / CGPA: 3.95/4.00

Dhaka, Bangladesh

2026

Thesis - Real-time Face Recognition System Leveraging Facenet and VGGFace Fusion embedding, and Siamese Network with ArcFace and Triplet Loss Function

Rajshahi University of Engineering & Technology

B.Sc. in Electronics & Telecommunication Engineering / CGPA: 3.04/4.00

Rajshahi, Bangladesh

2013 – 2018

Thesis - Moving object detection in video surveillance

Selected Research Publications

Research spanning applied ML, computer vision interpretability, federated systems, and predictive modelling & published across IEEE, ICECCME, ICCIT, and Zeitschrift für Naturforschung A.

- Performance-based comparative analysis of fine-tuned transfer learning approaches with Grad-CAM interpretability for dental caries prediction.
- Symptomatic & Non-Symptomatic Hepatocellular Carcinoma Prediction using Machine Learning, ICECCME.
- Prediction on Intent Classification of Java and C# Web Queries using Semi-supervision, ICCIT.
- A Horizontal Federated Random Forest for Heart Disease Detection from Decentralized Local Data, IEEE R10 HTC.
- ML Classifier Comparative Performance Analysis of Prediction on Cervical Cancer (ICECIT).
- Comparative Numerical Study Of Hole Transport Layer To Improve The Performance Of Cs₂TiI₆ Based Perovskite Solar Cell (EICT)
- Numerical study of highly efficient tin-based perovskite solar cell with MoS₂ hole transport layer (Journal- Zeitschrift für Naturforschung A)